

First Task

Principal components

The first task of the report is to analyse the crop yield data. In this data, we have 23 predictors and 1500 samples. However, after performing principal components analysis (PCA), it is known that only 11 principal components are needed to capture at least 80 percent of the variance in the data, as shown in Figure 1.

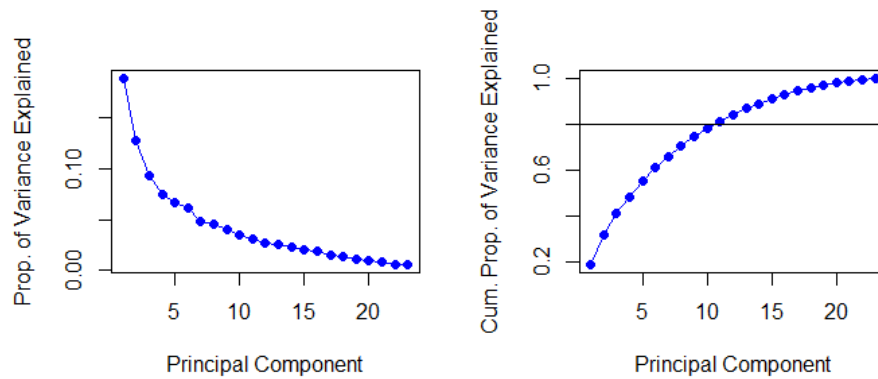


Figure 1. Left: scree plot depicting the proportion of variance explained by each of the principal components. Right: cumulative proportion of variance explained by the principal components.

Independent and identically distributed data check

To analyse whether the data is independent and identically distributed (i.i.d), several tests are performed. First, to evaluate the identity of the mean, ANOVA is performed both temporally and spatially. The test clearly shows that the data is neither temporally nor spatially identically distributed because both p-values are extremely low (2×10^{-16}).

Secondly, auto correlation function plot is used to check the temporal independence for each location. As shown in Figure 2, in almost all locations, the lag bars stay inside the statistically insignificant boundary lines. It means that the data is temporally independent.

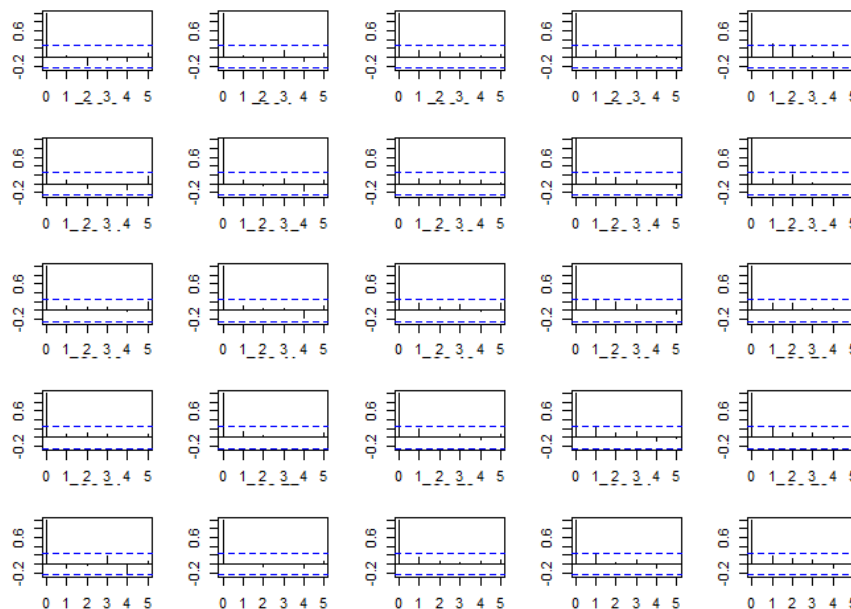


Figure 2. Auto correlation functions with 0 – 5 year lags in 25 locations

Finally, correlation matrix is used to see the independence between locations. As seen in Figure 3, most of the correlation blocks value is higher than 0.7, meaning that the data is dependent between each other. Interpretably, there is a high possibility that these locations are nearly situated or located in similar characteristic locations. Based on the performed tests, the data is not i.i.d.

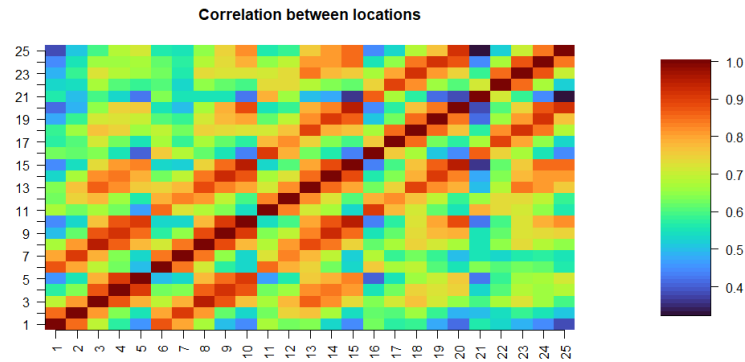


Figure 3. Correlation matrix of all 25 locations

Best subset regression and lasso regression model

Randomly split

The data is then randomly split with seed=99 into an equally sized training and validation set to perform best subset regression and lasso regression. Using the best subset regression method, the training data is trained. The MSE and R^2 are shown in Figure 4. Using this method, 22 variables are selected (all, except: temp_1) based on its lowest validation (test) MSE and R^2 values, shown in Figure 5.

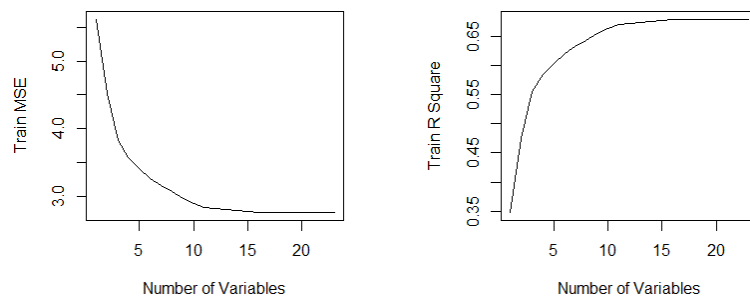


Figure 4. Left: train data MSE using random split. Right: train data R^2 using random split

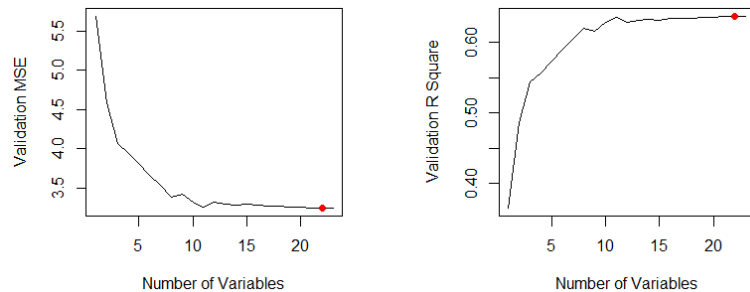


Figure 5. Left: test data MSE using random split. Right: test data R^2 using random split

Although the cross-validation index tracking estimates 20 variables during the folding process (shown in Figure 6), the final model coefficient matrix fitted on the full dataset yields 22 variables (all, except: min_temp)

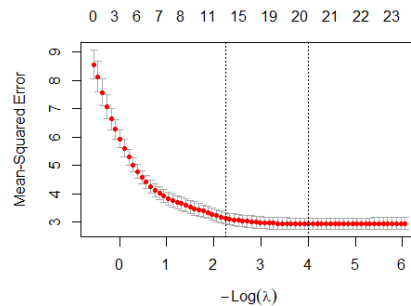


Figure 6. Lasso cross-validation plot using random split

Predicted and observed data is then plotted for both methods to check the performance. The plot shows a similar performance between these 2 methods.

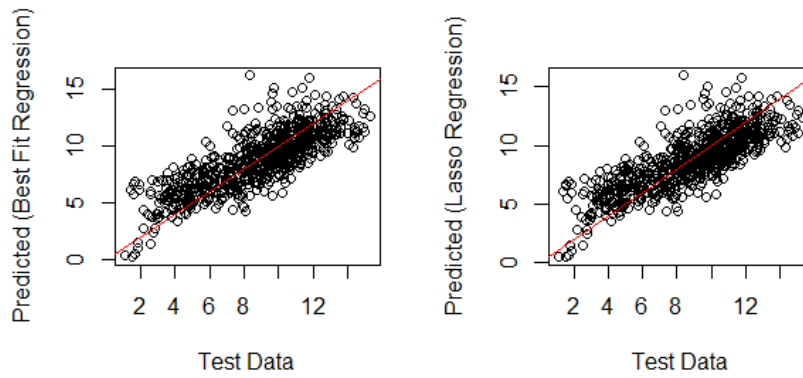


Figure 7. Predicted vs observed yields using random split

Even-uneven years split

The data is then splitted based on even and uneven years. Using this splitting, the selected variables in the best fit regression method are 12 variables (pr_2, pr_3, rad_2, rad_3, temp_2, temp_3, temp_4, temp_5, temp_6, warm_day, max_temp, max_5day_pr) with the training and test MSE and R^2 showed in Figure 8 and Figure 9.

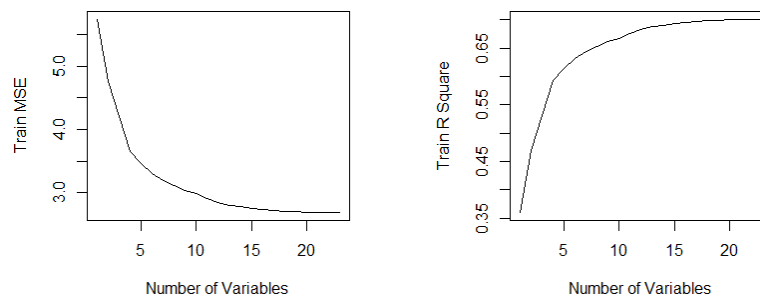


Figure 8. Left: Train data MSE using even-uneven splitting. Right: Train data R^2 using even-uneven split

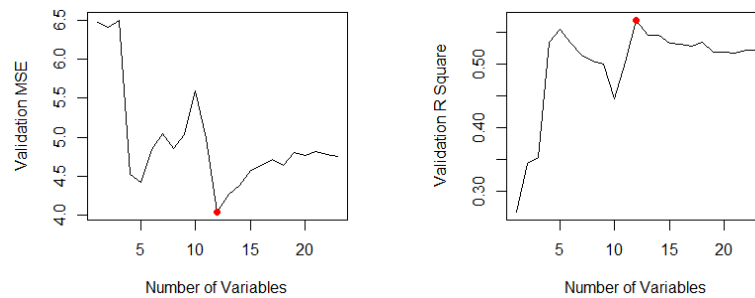


Figure 9. Left: test data MSE using even-uneven splitting. Right: test data R^2 using even-uneven split

The final model coefficient matrix fitted on the full dataset yields all variables (23 variables) as shown in Figure 10.

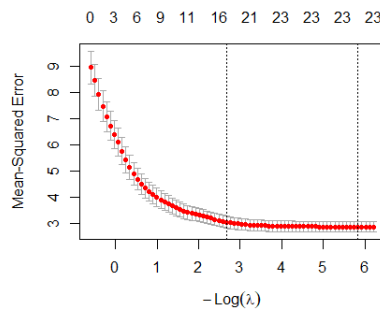


Figure 10. Lasso cross-validation plot using even-uneven split

Predicted and observed data is then plotted for both methods to check the performance in the even-uneven split data. The performance is considered lower than the random split, as the data is more scattered in the observed high yields area.

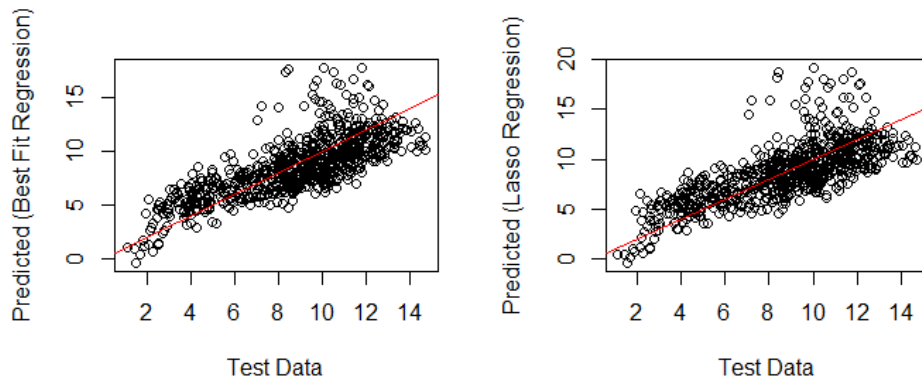


Figure 11. Predicted vs observed yields using even-uneven split

Selected variables, MSE, and R^2

Using best fit regression, the selected variables using even-uneven split method is considerably much lower compared to the randomly split data, although a higher MSE and lower R^2 has resulted as the trade off, as shown in Table 1.

Table 1. Selected variables, MSE, and R^2 comparison using best fit regression for random and even-uneven split

	Selected variables	Training		Test	
		MSE	R^2	MSE	R^2
Random split	22	2.75486	0.68011	3.24081	0.63623
Even-uneven split	12	2.84900	0.68232	4.03416	0.52810

Using Lasso regression, both splitting methods result a similar and large selected variables, as shown in Table 2

Table 2. Lambda, selected variables, MSE, and R^2 comparison using Lasso regression for random and even-uneven split

	Best lambda	Selected variables	Training		Test	
			MSE	R^2	MSE	R^2
Random split	0.01811	22	2.76701	0.67870	3.23733	0.63662
Even-uneven split	0.00293	23	2.68394	0.70072	4.71135	0.44889

Result discussion

Based on these results, best fit regression method combined with even-uneven splitting method is preferred in this specific case. The reason for this is due to the temporal data leakage in random splitting, where the test set contains information that is similar to the training set. Therefore, the MSE will continue to reduce with the addition of the variables. In even-uneven split, training and test data set is strictly temporally separated. This splitting method however does not work effectively using Lasso, due to the uniform shrinkage trade off by the continuous penalty. Lasso also tends to distribute weight across correlated predictor groups rather than strictly eliminating redundant features.

Second Task

The second task is to analyse the climate data. Out of 25600 data points, there are only 12041 non-NA data points. Based on this data, the lowest annual temperature is -6.98474°C and the highest is 24.7285°C . On the other hand, the lowest annual precipitation is 0 mm and the highest is 4938.4 mm. After fitting the data, it is known that the strongest temperature trend (slope) is 0.008634239, while the strongest precipitation trend is 0.0733172. Figure 12 shows all the trends at all locations.

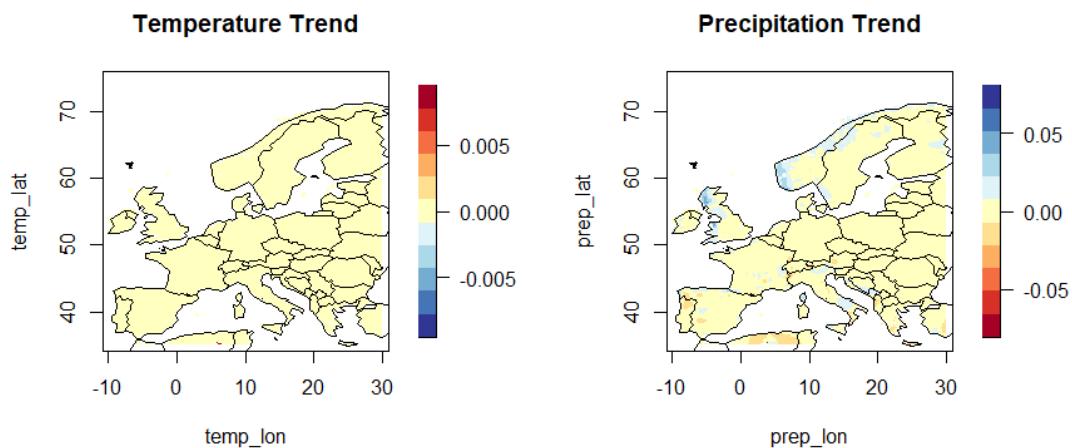


Figure 12. Left: temperature trend map at all locations. Right: precipitation trend map at all locations

The null hypothesis of the single testing is that there is no trend of temperature or there is no trend of precipitation at the specific locations between years. However, because we evaluated in many locations (multiple testing), false discovery rate is needed to be determined. Here, Benjamini-Hochberg (BH) is used as the method. Comparing single test with $\alpha=0.05$ and BH multiple test using $q=0.1$, the number of significant locations is 44 more in temperature data and 247 less in precipitation data in the multiple test, as shown in Table 3.

Table 3. Number of significant locations

	Single Testing ($\alpha=0.05$)	Multiple Testing (BH, $q=0.1$)
Temperature data	11734	11778
Precipitation data	4880	4633

The trend is then once again plotted on the map in Figure 13. It shows however only the locations that have significant trend based on the BH method. Based on this map, significant temperature trend appears in almost all Europe territory, while the significant precipitation trend concentrated only in northern Europe.

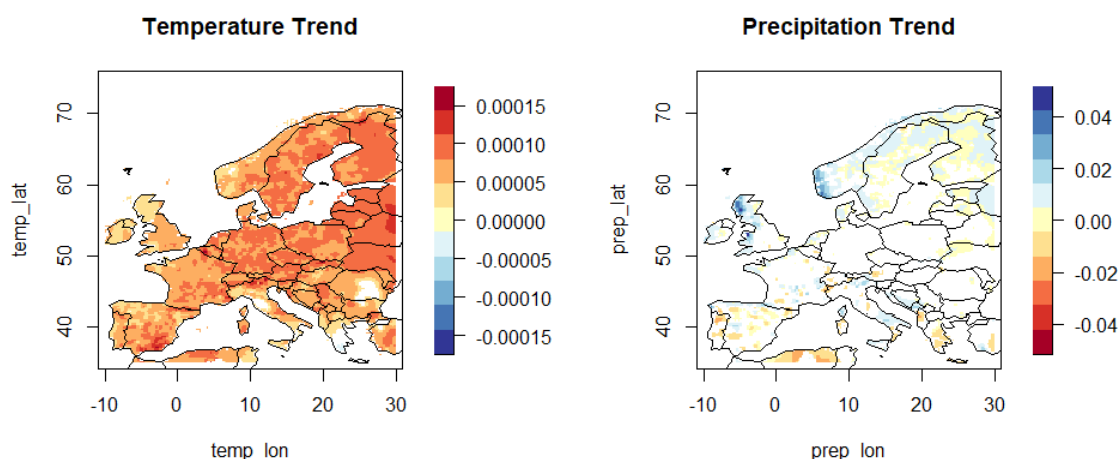


Figure 13. Left: temperature trend map at locations where the trends are significant. Right: precipitation trend map at locations where the trends are significant.

Comparing the false positives, the expected false positives look like almost doubled using BH in temperature data but decreased in the precipitation data as shown in Table 4. However, the expected positive values stated in multiple testing is the maximum value. It does not mean that the single testing is more preferred, because the multiple testing can guarantee that 90% of the significant data is valid significant trends, while the single testing cannot guarantee this. The BH multiple test has also a stronger power to identify trends, compared to the single testing. Therefore, it is preferred to use multiple testing method when a spatial data is being analysed.

Table 4. Expected false positives for single testing and maximum expected false positives for multiple testing

	Single Testing ($\alpha=0.05$)	Multiple Testing (BH, $q=0.1$)
Temperature data	602.5	1177.8
Precipitation data	602.5	463.3

Result discussion

Based on the result, the significant locations are increased using multiple testing, it is interpreted, that the temperature trend signal is strong, dominant, and spread in the majority of the locations. Unlike the temperature data, the precipitation data has high noise and the signal is not spread as large as the temperature data.